

2020 年秋季概率统计 C 考试答案(2021-1-3)

一. 1.D, 2.A, 3.B, 4.C, 5.D,

二. 1. $\frac{5}{12}$, 2. $f_Y(y) = \frac{2e^y}{\pi(1+e^{2y})}$, 3. $\geq \frac{11}{12}$, 4. $F(1,3)$, 5. (17.02,18.98),

三. 解: 设 $B =$ (从乙袋中取到白球), $A_1 =$ (从甲袋中取到两个白球),

$A_2 =$ (从甲袋中取到一个白球和一个红球), $A_3 =$ (从甲袋中取到两个红球),

$$(1) P(B) = \sum_{i=1}^3 P(A_i)P(B|A_i) = \frac{1}{6} \times \frac{7}{11} + \frac{5}{9} \times \frac{6}{11} + \frac{5}{18} \times \frac{5}{11} = \frac{53}{99} \quad 4, \text{ 6 分}$$

$$(2) P(A_1|\bar{B}) = \frac{P(A_1)P(\bar{B}|A_1)}{1-P(B)} = \frac{\frac{1}{6} \times \frac{4}{11}}{\frac{46}{99}} = \frac{3}{23} \quad 3 \text{ 分}$$

四. 解: (1) $f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \begin{cases} e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad 3 \text{ 分}$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} e^{x-y}, & 0 < x < y \\ 0, & \text{其他} \end{cases} \quad 5 \text{ 分}$$

(2) 其一: $Z = \max\{X,Y\} = Y$,

$$\text{故 } f_Z(z) = f_Y(z) = \int_{-\infty}^{+\infty} f(x,z)dx = \begin{cases} ze^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases} \quad 4 \text{ 分}$$

其二:

$$F_Z(z) = P(X \leq z, Y \leq z) = \begin{cases} \int_0^z dy \int_0^y e^{-y} dx = \int_0^z ye^{-y} dy = 1 - e^{-z} - ze^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases} \quad 2 \text{ 分}$$

$$\text{故 } f_Z(z) = F'_Z(z) = \begin{cases} ze^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases} \quad 4 \text{ 分}$$

五. 解: (1) 其一: $f_x(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其他} \end{cases}$ $f_z(z) = \int_{-\infty}^{+\infty} f_x(z-y)f_y(y)dy$ 2分

$$f_x(z-y)f_y(y)=\begin{cases}\frac{2y}{\pi^2}, & 0 < z < 1, 0 < y < z \\ \frac{2y}{\pi^2}, & 1 \leq z < \pi, z-1 < y < z \\ \frac{2y}{\pi^2}, & \pi \leq z < \pi+1, z-1 < y < \pi \\ 0, & \text{其他}\end{cases}$$

$$f_z(z)=\begin{cases}\frac{z^2}{\pi^2}, & 0 < z < 1 \\ \frac{2z-1}{\pi^2}, & 1 \leq z < \pi \\ 1-\frac{(z-1)^2}{\pi^2}, & \pi \leq z < \pi+1 \\ 0, & \text{其他}\end{cases}$$

7分

其二: $F_z(z) = \begin{cases} 0, & z < 0 \\ \frac{z^3}{3\pi^2}, & 0 \leq z < 1 \\ \frac{1}{\pi^2}(z^2 - z + \frac{1}{3}), & 1 \leq z < \pi \\ -\frac{(z-1)^3}{3\pi^2} + z - \frac{\pi}{3}, & \pi \leq z < \pi+1 \\ 1, & z \geq \pi+1 \end{cases}$ 4分

$$f_z(z)=F'_z(z)=\begin{cases}\frac{z^2}{\pi^2}, & 0 < z < 1 \\ \frac{2z-1}{\pi^2}, & 1 \leq z < \pi \\ 1-\frac{(z-1)^2}{\pi^2}, & \pi \leq z < \pi+1 \\ 0, & \text{其他}\end{cases}$$

7分

(2)其一: $E(Y) = \int_0^\pi \frac{2y^2}{\pi^2} dy = \frac{2\pi}{3}$, $E(U+V) = E(X+Y) = E(X) + E(Y) = \frac{1}{2} + \frac{2\pi}{3}$ 2分

其二: $F_Y(y) = \begin{cases} 0, & y \leq 0 \\ \frac{y^2}{\pi^2}, & 0 < y < \pi \\ 1, & y \geq \pi \end{cases}$

$$f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其他} \end{cases} \quad F_X(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$F_U(u) = P(X \leq u, Y \leq u) = P(X \leq u)P(Y \leq u) = F_X(u)F_Y(u)$$

$$f_U(u) = F'_U(u) = f_X(u)F_Y(u) + F_X(u)f_Y(u) = \begin{cases} \frac{3u^2}{\pi^2}, & 0 < u < 1 \\ \frac{2u}{\pi^2}, & 1 \leq u < \pi \\ 0, & \text{其他} \end{cases}$$

$$E(U) = \int_0^1 \frac{3u^3}{\pi^2} du + \int_1^\pi \frac{2u^2}{\pi^2} du = \frac{2\pi}{3} + \frac{1}{12\pi^2}$$

$$F_V(v) = 1 - P(X > vP(Y > v)) = 1 - (1 - F_X(v))(1 - F_Y(v))$$

$$f_V(v) = F'_V(v) = f_X(v)(1 - F_Y(v)) + (1 - F_X(v))f_Y(v) = \begin{cases} 1 + \frac{2v}{\pi^2} - \frac{3v^2}{\pi^2}, & 0 < v < 1 \\ 0, & \text{其他} \end{cases}$$

$$E(V) = \int_0^1 v(1 + \frac{2v}{\pi^2} - \frac{3v^2}{\pi^2}) dv = \frac{1}{2} - \frac{1}{12\pi^2}$$

$$E(U + V) = E(U) + E(V) = \frac{2\pi}{3} + \frac{1}{2} \quad 2 \text{ 分}$$

六. 解: $f(x; \theta) = \begin{cases} \frac{x}{\theta} e^{-\frac{x^2}{2\theta^2}}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad 1 \text{ 分}$

(1) ① $E(X) = \int_{-\infty}^{+\infty} xf(x)dx = \int_0^{+\infty} x \frac{x}{\theta} e^{-\frac{x^2}{2\theta^2}} dx = \frac{\sqrt{2\pi}\theta}{2} = \sqrt{\frac{\pi\theta}{2}} \quad 2 \text{ 分}$

$\theta = \frac{2}{\pi} [E(X)]^2 \quad \hat{\theta}_1 = \frac{2}{\pi} (\bar{X})^2 \quad 3 \text{ 分}$

② $L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \prod_{i=1}^n (\frac{x_i}{\theta} e^{-\frac{x_i^2}{2\theta^2}}) = (\prod_{i=1}^n x_i) \cdot \theta^{-n} \cdot e^{-\frac{\sum_{i=1}^n x_i^2}{2\theta^2}}, \quad x_i > 0 \quad 4 \text{ 分}$

$$\ln L(\theta) = \sum_{i=1}^n \ln x_i - n \ln \theta - \frac{\sum_{i=1}^n x_i^2}{2\theta}$$

$$\frac{d \ln L(\theta)}{d \theta} = -\frac{n}{\theta} + \frac{\sum_{i=1}^n x_i^2}{2\theta^2} = 0 \quad 5 \text{ 分}$$

$$\hat{\theta}_2 = \frac{1}{2n} \sum_{i=1}^n X_i^2$$

$$(2) \quad ① E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^{+\infty} x^2 \frac{x}{\theta} e^{-\frac{x^2}{2\theta}} dx = 2\theta, \quad D(X) = 2\theta - \frac{\pi\theta}{2}$$

$$E(\hat{\theta}_1) = \frac{2}{\pi} E(\bar{X}^2) = \frac{2}{\pi} \{D(\bar{X}) + [E(\bar{X})]^2\} = \frac{2}{\pi} \left\{ \frac{D(X)}{n} + [E(X)]^2 \right\} = \frac{4 - \pi + n\pi}{n\pi} \theta \neq \theta \quad 1 \text{ 分}$$

故 $\hat{\theta}_1$ 为 θ 的有偏估计

$$\text{修正 } \hat{\theta}_1 \text{ 为 } \hat{\theta}_3 = \frac{n\pi}{4 - \pi + n\pi} \hat{\theta}_1, \text{ 而 } \hat{\theta}_3 \text{ 为 } \theta \text{ 的无偏估计} \quad 2 \text{ 分}$$

$$② E(\hat{\theta}_2) = \frac{1}{2n} \sum_{i=1}^n E(X_i^2) = \frac{1}{2} E(X^2) = \frac{1}{2} \int_{-\infty}^{+\infty} x^2 f(x) dx = \frac{1}{2} \int_0^{+\infty} x^2 \frac{x}{\theta} e^{-\frac{x^2}{2\theta}} dx = \theta$$

故 $\hat{\theta}_2$ 为 θ 的无偏估计 3 分

$$(3) \quad \hat{\theta}_2 \xrightarrow{P} \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{2} E(X_i^2) \right] = \frac{1}{2} E(X^2) = \frac{1}{2} \int_0^{+\infty} x^2 \frac{x}{\theta} e^{-\frac{x^2}{2\theta}} dx = \theta, (n \rightarrow \infty)$$

故 $\hat{\theta}_2$ 为 θ 的相合估计。 1 分

七. 解: $F_z(z) = P(Z \leq z) = P(X + Y \leq z) =$ 1 分

$$= P(X=0)P(Y \leq z) + P(X=1)P(Y \leq z-1) + P(X=2)P(Y \leq z-2)$$

$$F_Y(y) = \begin{cases} 0, & y \leq 0 \\ y^2, & 0 < y < 1 \\ 1, & y \geq 1 \end{cases} \quad 2 \text{ 分}$$

$$F_Z(z) = \begin{cases} 0, & z < 0 \\ \frac{1}{4}z^2, & 0 \leq z < 1 \\ \frac{1}{4} + \frac{1}{2}(z-1)^2, & 1 \leq z < 2 \\ \frac{3}{4} + \frac{1}{4}(z-2)^2, & 2 \leq z < 3 \\ 1, & z \geq 3 \end{cases} \quad 4 \text{ 分}$$